

FOURTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2021

Mathematics

MEC 4C 04—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 4C 04—MATHEMATICAL ECONOMICS
(Multiple Choice Questions for SDE Candidates)

1. Regression analysis is concerned with the study of the dependence of :
 - (A) Explanatory variables on one or more dependent variables.
 - (B) Dependent variable on one or more explanatory variables.
 - (C) Both explanatory and dependent variables on other known variables.
 - (D) Two known variables.
2. Stochastic variables are :
 - (A) Deterministic values.
 - (B) Non-random variables.
 - (C) Imply causation.
 - (D) Have probability distribution.
3. Firm data collected for top 10 companies classified based on profitability for 10 years is an example of :
 - (A) Cross-sectional data.
 - (B) Time series data.
 - (C) Pooled data.
 - (D) Panel data.
4. The data on GDP, unemployment, household expenditure are examples of :
 - (A) Experimental data.
 - (B) Non-experimental data.
 - (C) Cross-section data.
 - (D) Time series data.
5. Variables such as grades in mathematics, results of horse race, degree of satisfaction at a restaurant are examples of :
 - (A) Ratio scale.
 - (B) Interval scale.
 - (C) Ordinal scale.
 - (D) Nominal scale.
6. The sample Regression line is at best an approximation of the true population regression. The statement :
 - (A) Is always true.
 - (B) Is always false.
 - (C) May sometimes be true sometimes false.
 - (D) Non-sense statement.
7. $Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i$:
 - (A) Sample regression function.
 - (B) Population regression function.
 - (C) Non-linear regression function.
 - (D) Estimate of regression function.

8. In $Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i$, $\hat{u}_i$ represent :
- (A) Fixed component. (B) Residual component.
(C) Estimates. (D) Estimators.
9. The regression model includes a random error or disturbance term for a variety of reasons. Which of the following is NOT one of them ?
- (A) Individual Y observations are intrinsically random even if they are measured correctly.
(B) Influence of variables other than X (Omitted variables).
(C) Unavailability of measurable data based on theory.
(D) Approximation errors in the calculation of the least squares estimates.
10. In the simple linear regression model, the regression slope :
- (A) Indicates by how many percent Y increases, given a one percent increase in X.
(B) When multiplied with the explanatory variable will give you the predicted Y.
(C) Indicates by how many units Y increases, given a one unit increase in X.
(D) Represents the elasticity of Y on X.
11. The statement that -There can be more than one SRF representing a population regression function is :
- (A) Always true. (B) Always false.
(C) Sometimes true, sometimes false. (D) Non-sense statement.
12. The population regression function is not directly observable. This is a :
- (A) True statement.
(B) False statement.
(C) Mostly true statement depending on the population.
(D) Mostly false statement depending on the observation capacity of researcher.
13. In $Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i$, $\hat{u}_i$ gives the differences between :
- (A) The actual and estimated Y values.
(B) The actual and estimated X values.
(C) The actual and estimated beta values.
(D) The actual and estimated u values.

14. Under the least square procedure, larger the $\hat{u}_i$ (in absolute terms), the larger the :
- (A) Standard error.
 - (B) Regression error.
 - (C) Squared sum of residuals.
 - (D) Difference between true parameter and estimated parameter.
15. Under normality assumption of u_i , the OLS estimator are :
- (A) Minimum variance unbiased.
 - (B) Consistent.
 - (C) $\hat{\beta}_1$ is normally distributed.
 - (D) All the above.
16. Rejecting a true hypothesis results in this type of error :
- (A) Type I error.
 - (B) Type II error.
 - (C) Structural error.
 - (D) Hypothesis error.
17. Accepting a false hypothesis results in this type of error :
- (A) Type I error.
 - (B) Type II error.
 - (C) Structural error.
 - (D) Hypothesis error.
18. The α in a confidence interval given by $\Pr(\hat{\beta}_2 - \delta \leq \beta_2 \leq \hat{\beta}_2 + \delta) = 1 - \alpha$ is known as :
- (A) Confidence coefficient.
 - (B) Level of confidence.
 - (C) Level of significance.
 - (D) Significance coefficient.
19. The α in a confidence interval given by $\Pr(\hat{\beta}_2 - \delta \leq \beta_2 \leq \hat{\beta}_2 + \delta) = 1 - \alpha$ should be :
- (A) < 0 .
 - (B) > 0 .
 - (C) < 1 .
 - (D) > 0 and < 1 .
20. In confidence interval estimation, $\alpha = 5\%$, this means that this interval includes the true β with probability of :
- (A) 5 %.
 - (B) 50 %.
 - (C) 95 %.
 - (D) 45 %.

**FOURTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

MEC 4C 04—MATHEMATICAL ECONOMICS

Time : Two Hours

Maximum : 40 Marks

Section A

Answer at least eight questions.

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

1. Distinguish between theoretical econometrics and applied econometrics
2. What are the goals of Econometrics ?
3. Distinguish between two variable regression and multiple regression analysis
4. Define Stochastic Disturbance term with stochastic specification of a model
5. What do you mean by standard error test ?
6. What are the properties of R^2 ?
7. Give a Ballentine view of $R^2 = 0$ and $R^2 = 1$.
8. What is Autocorrelation ?
9. What is confidence interval ?
10. Define reciprocal models with its functional form.
11. What do you mean by regression through the origin ?
12. What is forecasting or prediction ?

CHMK LIBRARY UNIVERSITY OF CALICUT

Page 1 of 1

Section B

*Answer at least **five** questions.*

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Explain different measurement scales of variables ?
14. Explain briefly statistical properties of OLS estimators.
15. Distinguish between PRF and SRF.
16. What is Maximum Likelihood Method (ML) ?
17. Explain Type I Error and Type II Error.
18. Why do we adjust R^2 ?
19. What do you mean by regression on standardized variables ?

(5 × 5 = 25 marks)

Section C

*Answer any **one** question.*

The question carries 11 marks.

20. What do you mean by regression ? Explain the method of Ordinary Least Squares.
21. Prove that OLS estimators are the best linear unbiased estimators (BLUE).

(1 × 11 = 11 marks)